Effect of Maternal Raspberry Leaf Consumption in Rats on Pregnancy Outcome and the Fertility of the Female Offspring

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Background. *The use of herbal medicines by pregnant women is on the rise. However, there is limited information regarding the safety of these compounds during pregnancy. Therefore, the goal of this study was to explore the consequences of raspberry leaf use during gestation in Wistar rats.* ***Methods.*** *Female rats were randomly assigned to receive vehicle, raspberry leaf, or specific flavonoids in raspberry leaf (kaempferol or quercetin; 10 mg/kg per day) orally once breeding had been confirmed until parturition. We assessed pregnancy outcomes in the P generation and reproductive development/fertility in the F1 raspberry leaf–exposed female offspring.* ***Results.*** *Raspberry leaf use during pregnancy was associated with increased gestation length and accelerated reproductive development in the F1 offspring.* ***Conclusions.*** *Results from this study have shown for the first time that raspberry leaf use during pregnancy can have long-term consequences for the health of the offspring and raise concerns about the safety of this herbal preparation for use during pregnancy.*

Key words: Raspberry leaf, intrauterine exposure, pregnancy loss, puberty, transgenerational effects.

INTRODUCTION

The use of herbal medicines has been rapidly increasing in North America in all segments of the population including pregnant women. Indeed, it has been estimated that as many as 60% of women use natural health products during their pregnancies.[1-4] Although much of the herbal consumption during pregnancy is accounted for by products that are used by the general population, such as *Echinacea*, gingko, and ginger,[5] there is a subset of preparations specifically targeted at pregnant women. Among these products, teas, capsules, and extracts made from or containing raspberry leaf are heavily promoted for use during all stages of pregnancy and lactation.

Widely disseminated information in the scientific and lay literature recommends the use of raspberry leaf tea, capsules, or infusions to prevent morning sickness.[6] It is also suggested that raspberry leaf tea helps the uterus work in a more efficient way during labor and birth, thus making birth easier and faster.[7] Indeed, Simpson et al[8] have reported that ingestion of raspberry leaf extract during the final weeks of pregnancy significantly shortened the second stage of labor. However, a recent survey of the popular literature found that raspberry leaf was commonly designated as unsafe for use in pregnancy and has been reported by some sources to be an abortifacient.[6] It is important to note that in many instances, sources recommending herbal use during pregnancy often do not cite scientific or clinical evidence to support these warnings; likely because to a large degree this information does not

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exist.[6] Furthermore, the long-term health consequences of fetal exposure to these herbal preparations have not been determined.

The goal of this study was therefore to explore the effects of exposure to raspberry leaf during gestation on pregnancy outcomes and postnatal reproductive outcomes in Wistar rats. In addition to examining the whole herbal preparation (ie, raspberry leaf), we examined whether specific flavonoids (ie, kaempferol and quercetin) present in raspberry leaf could affect these outcomes in isolation.

MATERIALS AND METHODS

Maintenance and Treatment of Animals

All animal experiments were approved by the Animal Research Ethics Board at McMaster University, in accordance with the guidelines of the Canadian Council for Animal Care. Nulliparous 200 to 250 g female Wistar rats (Harlan, Indianapolis, Ind) were maintained under controlled lighting (12:12 L:D) and temperature (22°C) with ad libitum access to food and water. Dams were randomly assigned to receive vehicle, 10 mg/kg per day red raspberry leaf (*Rubus idaeus*; Nature's Way, Canmore, Alberta, Canada), 10 mg/kg quercetin monohydrate (Sigma Aldrich, Oakville, Ontario, Canada) or 10 mg/kg kaempferol (TCI America, Portland, Ore) orally in a flavored gelatine base[9] once breeding had been confirmed by the presence of sperm in the vaginal swab (n = 10 per group). The doses of the test compounds were chosen based on studies which have shown that quercetin at this concentration is effective in rodent studies.[10,11] The day that a positive sign of copulation was observed was designated gestational day 0 (GD0). Treatments were administered from GD0 until the day of parturition. For each dam, gestation length, litter size, litter weight, birthweight, sex, and the number of stillbirths were recorded. From these data, the pregnancy success rate (*[# of dams delivering a litter/# dams with a confirmed mating]*100*), the live birth index (*[# of live offspring/# of offspring delivered]*100*), and the sex ratio (*# of male offspring/# of female offspring*) were calculated. The offspring were weighed weekly until weaning (postnatal day 21; PND21). For each litter, survival to PND4 and survival to weaning were determined. At weaning (PND21), female offspring were randomly selected, housed as sibling pairs, and checked daily for vaginal opening, an estimate of the onset of puberty, to determine whether fetal exposure to raspberry leaf could affect the onset of sexual maturity.

Fertility Assessment in F1 Female Offspring

Fertility in the female offspring of the control and raspberry leaf–treated dams was assessed at 6 months of age. Female offspring (F1; control N = 6, raspberry N = 8) were housed 1:1 with a proven male and monitored daily for confirmation of breeding (ie, the presence of sperm in the vaginal swab). The day that a positive sign of copulation was observed was designated gestational day 0 (GD0). For each dam (F1), time to pregnancy (ie, days until detection of sperm in the vaginal swab), gestation length, litter size, litter weight, birthweight, sex, and the number of stillbirths were recorded. Mating success *([# of females mated/# of females cohabited]*100)*, pregnancy success rate, live birth index, sex ratio, survival to PND4, and weaning were also determined.

Statistical Analysis

All statistical analyses were performed by Student *t* test (SigmaStat, v.2.03; SPSS, Chicago, Ill) comparing the treatment group to the controls using the litter as the experimental unit. Data were tested for normality as well as equal variance, and when normality or variance tests failed, data were analyzed using the Mann-Whitney rank sum test ($\alpha = .05$). Categorical variables were compared using Fisher's exact test ($\alpha = .05$). Values are presented as mean $\pm$ SEM.

RESULTS

Pregnancy and Birth Outcomes (P Generation)

There was no effect of maternal raspberry leaf exposure on maternal weight gain during pregnancy, the live birth index, litter size, birthweight, total litter weight, sex ratio, survival to PND4, or survival to weaning (Table 1). However, raspberry leaf exposure during pregnancy significantly increased gestation length (Table 1) and also appeared to be associated with reduced pregnancy success. In control animals, 100% of the dams with a confirmed mating delivered a litter, whereas only 78% of the raspberry leaf–treated dams with a confirmed mating had a full-term pregnancy. However, this difference did not reach statistical significance ($P > .05$). Raspberry leaf exposure during pregnancy resulted in precocious puberty (ie, advanced vaginal opening) in the female offspring (Figure 1).

Table 1. Effect of Raspberry Leaf and Selected Flavonoid Consumption on Pregnancy Outcomes[a]

P Fertility	Control, N = 10	Raspberry Leaf, N = 10	Kaempferol, N = 10	Quercetin, N = 10
Pregnancy weight gain (g)	150 ± 15.7	157 ± 7.65	159 ± 6.64	198 ± 13.46*
Gestation length (days)	20.8 ± 0.63	22.4 ± 0.37*	21.8 ± 0.22	21.3 ± 0.24
Pregnancy success rate (%)	100	78	100	90
Live birth index (%)	99.4 ± 0.60	99.0 ± 1.02	100	99.2 ± 0.80
Litter size (n)	12.2 ± 1.50	12.4 ± 1.07	15.1 ± 1.07	16.0 ± 0.96
Birth weight (g)	6.2 ± 0.21	6.9 ± 0.35	5.9 ± 0.17	5.9 ± 0.17
Total litter weight (g)	73.5 ± 8.07	82.7 ± 4.66	88.5 ± 5.40	93.3 ± 4.02
Sex ratio (M/F)	2.6 ± 1.00	1.0 ± 0.12	1.1 ± 0.22	1.0 ± 0.12
Survival to PND4 (%)	96.7 ± 3.33	98.2 ± 1.79	100	100
Survival to weaning (%)	96.7 ± 3.42	98.2 ± 1.79	100	98.6 ± 1.39

Abbreviations: F, female; M, male; PND4, postnatal day 4.

[a] Data are presented as mean ± SEM. Values marked with an asterisk (*) are significantly different ($P < .05$) from controls.

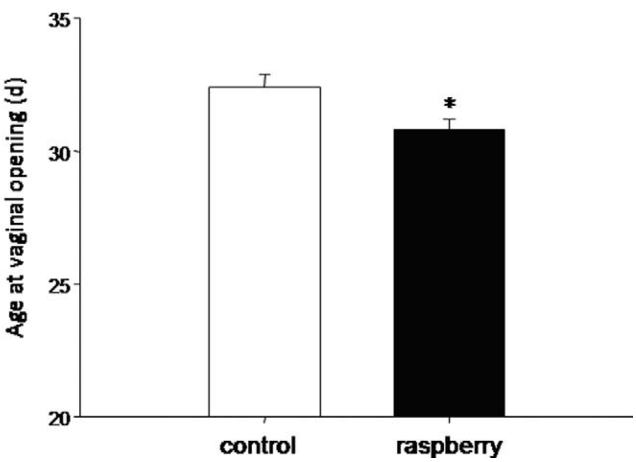

Figure 1. Time to vaginal opening in offspring exposed in utero to vehicle or raspberry leaf. Data are presented as mean ± SEM. An asterisk (*) indicates $P < .05$ versus vehicle control.

Dams given quercetin during pregnancy had significantly increased weight gain during pregnancy; however, there was no effect of either kaempferol or quercetin on any other pregnancy or birth outcome (Table 1).

Fertility of F1 Offspring

In the F1 offspring, there was no effect of fetal exposure to raspberry leaf on time to pregnancy, gestation length, mating success, pregnancy success, the live birth index, litter size, birthweight, total litter weight, sex ratio, or postnatal survival to PND4 or PND21 (Table 2). However, although average birthweight of the F2 pups was not significantly different between raspberry leaf–exposed and control dams, there was a significant ($P = .02$) increase in the proportion (9.8% vs. 0%; $P = .02$) of their offspring that were growth restricted (ie, had a birthweight that was more than 2 standard deviations below the mean birthweight of the control offspring).

DISCUSSION

In the lay press, the use of red raspberry leaf during pregnancy is often suggested for the treatment of morning sickness, to prevent miscarriage and preterm labor, to shorten labor, and to prevent postpartum hemorrhage and postmaturity.[5,12,13] Indeed, raspberry leaf is one of the most common herbal supplements taken during pregnancy.[13,14] More recently, there have been articles in the lay press claiming that raspberry leaf use during pregnancy can cause miscarriage. However, these articles are unaccompanied by any evidence to support this claim. In our study, only 78% of the raspberry leaf–exposed dams with a confirmed mating delivered a litter versus 100% of all control animals. Although this result was not statistically significant ($P > .05$), it may be biologically relevant and merits further investigation.

Although raspberry leaf contains numerous phytochemicals including tannins, phenolic acids, monoterpenes, and flavonoids,[15,16] the current literature suggests the active principle constituents affecting pregnancy success in this model are the flavonoids. The increased rate of miscarriage in humans taking raspberry leaf during pregnancy reported in the lay press may reflect impaired fertilization and/or reduced implantation, both of which have been shown to be affected by flavonoids. Exposing sperm to isolated flavonoids, including those found in

Table 2. Fertility Measures in Female Rats Exposed to Raspberry Leaf In Utero[a]

F1 Offspring Fertility	Control, N = 6	Raspberry Leaf, N = 8
Time to pregnancy (days)	3.2 ± 1.07	3.0 ± 0.49
Gestation length (days)	21.5 ± 0.35	21.8 ± 0.28
Mating success (%)	83	100
Pregnancy success rate (%)	83	87.5
Live birth index (%)	96.7 ± 1.90	99.2 ± 0.84
Litter size (n)	15.0 ± 0.91	15.6 ± 1.73
Birth weight (g)	6.2 ± 0.30	5.7 ± 0.16
Total litter weight (g)	89.0 ± 8.85	87.5 ± 7.00
Sex ratio (M/F)	0.7 ± 0.24	0.9 ± 0.12
Survival to PND4 (%)	93.8 ± 6.25	96.5 ± 2.62
Survival to weaning (%)	93.8 ± 6.25	86.6 ± 4.70

Abbreviations: F, female; M, male; PND4, postnatal day 4.
[a] Data are presented as mean ± SEM.

raspberry leaf, can result in premature activation of sperm and a decreased ability of sperm to penetrate cumulus cells.[17,18] In addition, the flavonoid quercetin has been shown to reduce sperm viability and motility in vitro.[19] It is possible that in our study an impairment in sperm function could have affected fertilization in the raspberry leaf-exposed dams. However, whether flavonoids in the female reproductive tract following oral administration of raspberry leaf reach concentrations that are high enough to affect sperm function is unknown. It has also been hypothesized that flavonoids are the active principle constituent of many plants with anti-implantation effects.[20] Indeed, the isolated flavonoids apigenin, luteolin, and daidzein have been shown to significantly impair implantation in rats.[21,22] Although isolated flavonoids can affect sperm function and implantation, the lack of effect of either quercetin or kaempferol on pregnancy success in this study suggests neither of these 2 raspberry leaf constituents is responsible for the reported increase in miscarriage associated with raspberry leaf exposure in humans. Further experiments are required to determine the mechanisms by which raspberry leaf can affect pregnancy success.

Raspberry leaf is commonly taken during pregnancy even though there is limited information available regarding the long-term health consequences of fetal exposure to this herbal preparation.[13,14] Results from this study have demonstrated that raspberry leaf exposure during pregnancy does have implications for reproductive health of the female offspring. Specifically, the offspring of raspberry leaf–exposed dams had earlier vaginal opening (ie, precocious puberty) relative to control offspring. One hypothesis for this acceleration of puberty in female offspring following raspberry leaf exposure is that it is due to the estrogenic actions of the flavonoid kaempferol present in raspberry leaf that has been shown to have estrogenic activity in MCF7 breast cancer cells.[23] Similarly, numerous animal studies have demonstrated that fetal exposure to flavonoids can accelerate vaginal opening[24]; however, neither kaempferol nor quercetin exposure alone affected the time to vaginal opening (data not shown). An alternate hypothesis is that the precocious puberty observed in the raspberry leaf–exposed offspring is a reflection of the increased body weight in these animals. Many human studies have shown a positive relationship between overweight/obesity and the onset of puberty in female children. In our study, although the raspberry leaf–exposed offspring tended to be heavier at birth ($P = .078$), the average body weight was the same as control animals at vaginal opening (control 113 ± 6.2 g vs. raspberry leaf–exposed 113 ± 2.1 g), even though this occurred approximately 2 days earlier. Although the exact mechanism by which fetal exposure to raspberry leaf results in accelerated onset of puberty cannot be determined at this time, it may have significant implications for children born to women who take this herbal preparation during pregnancy. Human studies have identified that girls who become sexually mature at a younger age are more likely to engage in risk-taking behaviors such as engaging in unprotected sex and may be at increased risk of reproductive cancers in later life.[25,26]

Furthermore, results from our study demonstrate that when the raspberry leaf–exposed F1 offspring were mated, the offspring were less healthy. Indeed, dams that were exposed to raspberry leaf in utero had a significantly higher proportion of offspring (F2) that were growth restricted, which is a known risk factor for obesity, type 2 diabetes, and hypertension in adult life.[27] Moreover, the proportion of dams with at least 1 pup that did not survive to weaning (PND21) was more than doubled in the raspberry leaf–exposed dams (57% vs. 25%), although this was not statistically significant. Taken together, these data suggest fetal exposure to raspberry leaf may have previously unidentified transgenerational effects that affect the health of the F2 offspring.

In summary, we have demonstrated for the first time that in Wistar rats, exposure to raspberry leaf extract throughout pregnancy increases gestation length and results in altered reproductive development and function in the offspring. There are some remarkable similarities between the effect of raspberry leaf consumption (the current study) and the effects of ginger tea consumption by rats during pregnancy.[28] Just as raspberry leaf exposure tended to reduce pregnancy success in our study, ginger tea consumption doubled embryonic loss when compared with controls.[28] Moreover, the trend toward increased birth weight in raspberry leaf-exposed offspring was paralleled by a significant increase in fetal weight in ginger-exposed animals.[28] Although the doses of herbals used in these studies are higher than what is considered normal human use, these studies suggest herbal use during pregnancy is not innocuous and further highlights the urgent need to evaluate the safety of phytomedicine use during pregnancy.

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